Software Documentation

1. Software Overview

- *What does the software do?*
  The App controls the turret system and the trigger by the user selecting six inputs: Left, Right, Up, Down, Trigger On, and Trigger Off.

  For model design, for both the Crossbow and the Target Box, I have used digital tools as mentioned in the "platforms used" section.
- *Platforms used (MicroPython, MIT App Inventor, etc.)*
  MicroPython, MIT App Inventor.

  For mode design, LibreCAD, Fusion 360, Blender, TinkerCAD.

2. System Logic

Explain flow in simple steps

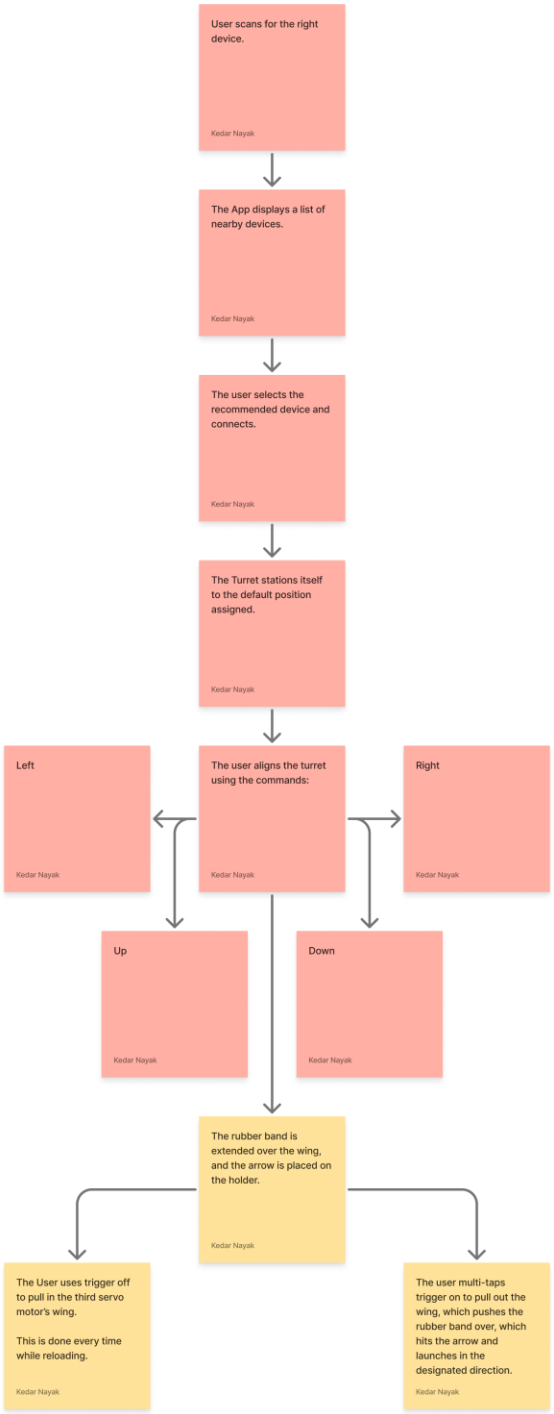
1. *User scans for the right device.*
2. *The App displays a list of nearby devices.*
3. *The user selects the recommended device and connects.*
4. *The Turret stations itself to the default position assigned.*
5. *The user aligns the turret using the commands: Left, Right, Up, and Down.*
6. *The User uses trigger off to pull in the third servo motor's wing.*
7. *The rubber band is extended over the wing, and the arrow is placed on the holder.*
8. *The user multi-taps trigger on to pull out the wing, which pushes the rubber band over, which hits the arrow and launches in the designated direction.*

Example:

1. *User presses button*
2. *App sends data via Bluetooth*
3. *ESP32 receives data*
4. *LED changes color*

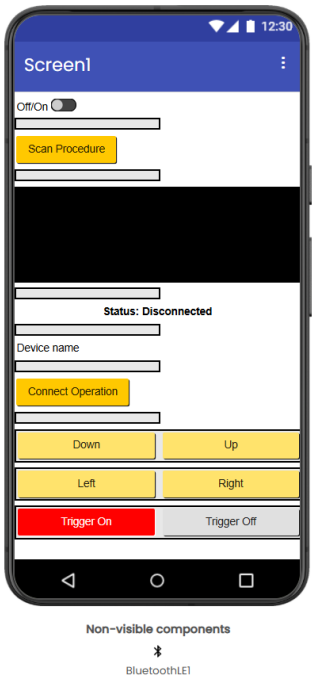
3. Flowchart

Visual representation of logic



4. App Flow (If MIT App Inventor)

- *Screen flow*



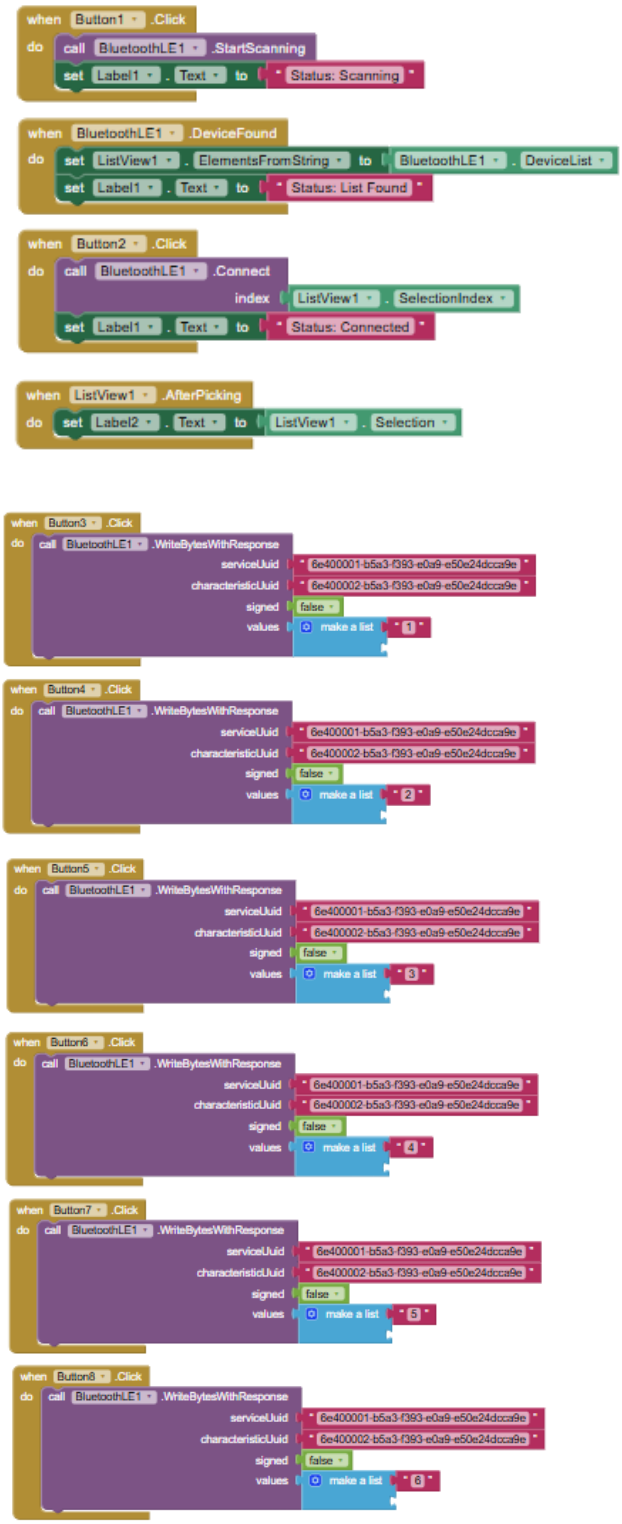
We have a fairly simple interface with direct, easy-to-understand instructions. The buttons are colour-coded with our model theme.

- *Button actions*
  The button "Scan Procedure" starts the scanning for nearby devices. (The black screen displays the list of devices. The user must select the required option)

  The button "Connect Operation" is clicked after the user chooses the necessary device, which connects to it.

  The four buttons in yellow are the alignment controls for the turret.

  The red trigger on and white trigger off are responsible for the launch and reloading phase of the Crossbow.

- *Data flow*
  After the connection is secured, every input the user gives is sent to the code, which processes the input to provide the stated output. Where each control button, especially the turret and trigger control, provides numbers from 1 to 6, each of which is received.

Each of these numbers is connected to a specific function, such as 1 to move the horizontal servo motor to the left one step at a time.